

CO1 ASSESSMENT TOOL3 OUTPUT PRODUCTION

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COURSE CODE – CSA0307

COURSE NAME – DATA STRUCTURES

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

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CO1-Assessment Tool- Output Prediction

QUESTIONS
Q1
Reverse a list
1 mark
Q2
Reverse a string
1 mark
Q3
Searching
1 mark
1/3 answered

Q1 Reverse a list 1 mark
Write a C program that takes an array as input and returns it reversed.
Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int arr[100], n, i;
5     printf("enter the number of element: ");
6     scanf("%d", &n);
7     printf("enter %d elements:\n", n);
8     for(i=0;i<n;i++) {
9         scanf("%d", &arr[i]);
10    }
11    printf("reversed array: ");
12    for(i=n-1;i>=0;i--) {
13        printf("%d", arr[i]);
14    }
15    return 0;
16 }
17
```

Run Save Submit

Input
5
10 20 30 40 50
Output
System error.

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CO1-Assessment Tool- Output Prediction

QUESTIONS
Q1
Reverse a list
1 mark
Q2
Reverse a string
1 mark
Q3
Searching
1 mark
1/3 answered

Q2 Reverse a string 1 mark
Write a C program for palindrom
Your Code

```
1 #include<stdio.h>
2 int main() {
3     int num, original, reverse = 0, remainder;
4     printf("enter a number: ");
5     scanf("%d", &num);
6     original= num;
7     while(num!=0) {
8         remainder=num%10;
9         reverse=reverse*10+remainder;
10        num=num/10;
11    }
12    if(original == reverse)
13        printf("%d is the pallindrom.\n", original);
14    else
15        printf("%d is not the pallindrom.\n", original);
16    return 0;
17 }
18 }
19
```

Run Save Submit

Input
121
Output
System error.

QUESTIONS

Q1

Reverse a list
1 mark

Q2

Reverse a string
1 mark

Q3

Searching
1 mark

1/3 answered

Q3 Searching

1 mark

Write a C program for Searching an Element in a Sorted Array

Your Code

```
1 #include<stdio.h>
2 int main() {
3     int arr[100], n, key;
4     int low, high, mid, found=0;
5     printf("enter the number of element:");
6     scanf("%d", &n);
7     printf("enter %d elements in sorted order:\n", n);
8     for(int i=0;i<n;i++) {
9         scanf("%d", &arr[i]);
10    }
11    printf("enter the element to search:");
12    scanf("%d", &key);
13    low=0;
14    high=n-1;
15    while(low<=high) {
16        mid=(low+high)/2;
17        if (arr[mid]==key) {
18            found=1;
19            break;
20        }
21        else if(arr[mid]<key) {
22            low=mid+1;
23        }
24        else {
25            high=mid-1;
```

Input

```
5
10 20 30 40 50
40
```

Output

System error.

1/3 answered

```
8     for(int i=0;i<n;i++) {
9         scanf("%d", &arr[i]);
10    }
11    printf("enter the element to search:");
12    scanf("%d", &key);
13    low=0;
14    high=n-1;
15    while(low<=high) {
16        mid=(low+high)/2;
17        if (arr[mid]==key) {
18            found=1;
19            break;
20        }
21        else if(arr[mid]<key) {
22            low=mid+1;
23        }
24        else {
25            high=mid-1;
26        }
27    }
28    if(found)
29        printf("element found at index %d.", mid);
30    else
31        printf("element not found.");
32    return 0;
33 }
```

System error.

